

# **AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA**

25-26J-479

## **Project Proposal Report**

Ridimahaliyadda A.K

B.Sc. (Hons) Degree in Information Technology Specialized in  
Software Engineering

Department of Computer Science and Software Engineering

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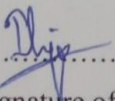
Sri Lanka

August 2025

## DECLARATION

I hereby declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher education, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name                | Student ID | Signature                                                                           |
|---------------------|------------|-------------------------------------------------------------------------------------|
| Ridimahaliyadda A.K | IT22250674 |  |

|                                                                                                                                |                                    |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <br>.....<br>Signature of the Supervisor     | <u>23/08/2025</u><br>.....<br>Date |
| <br>.....<br>Signature of the Co-Supervisor | <u>23/08/2025</u><br>.....<br>Date |

## **ABSTRACT**

The volatility of pepper price in Sri Lanka poses critical challenges to farmers, exporters, and policymakers, impacting decision-making and income stability. The goal of this research is to create an intelligent Market Forecasting Module that uses cutting-edge machine learning techniques to forecast weekly pepper prices. The research focusses on applying the Temporal Fusion Transformer (TFT) model for intricate, multivariate time-series forecasting tasks. This model incorporates a range of data sources, such as past prices, weather trends, exchange rates, and export volumes.

In parallel, the system incorporates a blockchain-based traceability mechanism to ensure data transparency, integrity, and trust across the pepper supply chain. Outputs from predictive models, including yield estimations, grading evaluations, and price forecasts, are securely recorded on a permission blockchain. A distinct QR code is then associated with each pepper batch, giving farmers, buyers, exporters, and regulators immediate access to tamper-proof, verifiable production and market data.

These approaches not only enhance forecasting accuracy and market readiness but also introduces a novel, farmer-focused digital infrastructure that ensures transparent, data-driven transactions and traceability. By aligning machine learning with blockchain technology, the proposed system offers a pioneering solution to current inefficiencies in Sri Lanka's pepper industry and sets the foundation for scalable, technology-driven agricultural supply chains. The proposed methodology and system design will be evaluated using real-world datasets and practical deployment frameworks. It is anticipated that the results will provide important stakeholders with useful information, equitable pricing policies, and confirmed product integrity.

**Key Words:** Pepper Price Forecasting, TFT, Blockchain Traceability

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## LIST OF ABBREVIATIONS

| Abbreviation | Description                                       |
|--------------|---------------------------------------------------|
| TFT          | Temporal Fusion Transformer                       |
| ML           | Machine Learning                                  |
| AI           | Artificial Intelligence                           |
| IoT          | Internet of Things                                |
| LSTM         | Long Short-Term Memory                            |
| ARIMA        | Autoregressive Integrated Moving Average          |
| SARIMA       | Seasonal Autoregressive Integrated Moving Average |
| AR           | Autoregressive Model                              |
| RF           | Random Forest                                     |
| TCN          | Temporal Convolutional Network                    |
| NDVI         | Normalized Difference Vegetation Index            |

# 1. INTRODUCTION

The Pepper Industry in Sri Lanka plays a crucial role in the nation's agricultural exports, yet it faces persistent challenges such as volatile market prices, lack of standardized quality grading, minimal transparency, and limited farmer-to-buyer traceability. Farmers are often disadvantaged by non-uniform pricing mechanisms where buyers set inconsistent rates with little correlation to production costs, resulting in revenue instability and exploitation. Additionally, there is no unified platform to integrate predictive market insights with verifiable product traceability, limiting the competitiveness of Sri Lankan pepper in global markets.

Price forecasting in the pepper industry has traditionally been carried out using statistical models such as ARIMA and SARIMA, which depend heavily on historical data and fail to account for external influences like weather conditions, global demand, and currency fluctuations. While more advanced machine learning methods such as Random Forest, XGBoost, and LSTM have improved prediction accuracy, they are often criticized for functioning as "black box" models with limited interpretability for end-users. To overcome these shortcomings, this research introduces the use of the TFT, a state-of-the-art deep learning architecture specifically designed for multi-horizon forecasting with interpretability. By combining numerical with unstructured data such as market news sentiment, TFT provides accurate forecasts while also highlighting which variables most influenced each prediction. This empowers farmers and policymakers with both accuracy and trust in the system's recommendations.

The Blockchain Integrity component addresses another critical issue in the pepper industry: lack of transparency and trust in the supply chain. Data generated from each module including yield predictions, grading assessments, and market price forecasts will be recorded on a permissioned blockchain, ensuring immutability, security, and tamper-proof traceability. Each batch of pepper will be linked to a unique QR code, allowing buyers and consumers to verify its origin, quality grade, production practices, and certification status in real time. This not only enhances consumer trust and supports fair trade practices but also gives Sri Lankan

pepper a stronger competitive edge in international markets, where transparency and traceability are increasingly demanded.

By integrating predictive analytics with blockchain-enabled traceability, this research goes beyond conventional isolated solutions and introduces a holistic decision support system for smart pepper farming. The combined approach empowers farmers to make informed decisions on when to sell, provides buyers with verified product data, and strengthens Sri Lanka's global market reputation through fairness, authenticity, and digital transparency. Furthermore, this system positions Sri Lankan pepper within the global paradigm of digital agriculture, aligning with international shifts toward sustainable farming, traceable supply chains, and AI-driven market insights. It bridges the gap between farmers, exporters, and consumers, offering a farmer-friendly yet globally competitive technological solution.

## **Sub-Topic: Market Price Prediction and Blockchain Integrity**

### **1.1 Background and Literature Review**

Price instability and opaque supply chains are two main challenges to Sri Lanka's pepper industry's long-term growth, as they directly reduce farmer income and export potential. Traditional statistical models, such as the ARIMA and its seasonal variation SARIMA, have been frequently utilized for agricultural commodity forecasting, including pepper, since they can simulate both seasonal patterns and linear trends using historical data [10]. A key drawback of these univariate approaches is that they solely rely on historical pricing data, neglecting the numerous exogenous factors such as weather anomalies, shifts in global demand, and trade policies that impact market swings. Consequently, they are often inadequate in periods of structural or market instability [11].

The advancement of ML has enabled more sophisticated forecasting algorithms that can manage multivariate, non-linear connections. Algorithms like RF and XGBoost have demonstrated improved forecast accuracy by integrating a range of information, including weather patterns, manufacturing production from competing nations, and macroeconomic indices [12]. Furthermore, deep learning architectures like LSTM networks, which excel at spotting complex temporal correlations in sequential data, enhance time-series forecasting [13]. Despite their advanced capabilities, these models' pervasive "black box" nature poses a significant barrier to their practical adoption in agricultural decision-making; stakeholders' inability to comprehend the logic behind predictions undermines trust and limits their applicability for risk management and strategic planning by farmers and exporters [14].

The TFT, a cutting-edge deep learning model created especially for multi-horizon time series forecasting, addresses this disparity between interpretability and accuracy [1]. In addition to its great precision, the TFT's architecture includes parts designed specifically for a variety of selection and interpretable self-attention processes. This allows it to pinpoint the precise factors that influence each forecast, such as a change in international trade laws or the

approaching monsoon. Actionable insights and increased user confidence are made possible by the capacity to visually explain projections, which aids stakeholders in understanding why a particular price change is anticipated [4]. The potential of such advanced models is further amplified in hybrid systems, as demonstrated by Zhao et al., where a TCN integrated with XGBoost showed remarkable results in forecasting agricultural price volatility by leveraging the complementary strengths of different algorithmic approaches [6].

Concurrently, the use of blockchain has emerged as a powerful potential solution to the problem of supply chain opacity in agriculture. Decentralization, immutability, and transparency are the fundamental features of blockchain that make it the perfect platform for developing tamper-proof digital ledgers to track product provenance from farm to fork. This is revolutionary for a high-value commodity like pepper, where ethical sourcing and authenticity are crucial [2]. When combined with IoT sensors for automated data collection such as temperature and humidity during transit and QR codes that give customers immediate, verifiable access to a product's complete history and journey, academic research has repeatedly confirmed the role of blockchain in improving food safety and traceability [7], [15].

Effective blockchain process that provide end-to-end traceability solutions are demonstrated in studies, guaranteeing that assertions regarding organic certification or place of origin are unquestionably recorded on an unchangeable ledger [8]. To improve the effectiveness and security of these traceability systems, additional innovations include the incorporation of RFID technology and cryptographic optimizations [16]. The effectiveness of such systems is not just technical; there is a substantial market opportunity for differentiated products because evidence indicates that consumers are becoming more willing to pay more for goods with a verifiable provenance [17]. Blockchain-based traceability is still largely neglected in Sri Lanka's spice industry, despite its demonstrated potential and increasing uptake in global commodity chains for goods like coffee and grains. This represents a substantial lost opportunity for value addition and market differentiation [3], [18].

A modern, complementary method of addressing both issues of integrity assurance and economic uncertainty is the combination of interpretable price forecasting through TFT with

blockchain-enabled traceability. The TFT component specifically addresses economic uncertainty by providing high-accuracy, multi-horizon price predictions for pepper markets while offering unprecedented interpretability through its attention mechanisms and variable selection networks [1]. This allows stakeholders to understand not just what the predicted price movement is, but which specific factors such as weather patterns, global demand shifts, or production forecasts from competing regions are driving each prediction, enabling more informed decision-making for planting, harvesting, and selling strategies [4]. When this interpretable forecasting capability is integrated with blockchain's immutable ledger system for tracking product provenance and quality metrics throughout the supply chain, the combined framework creates a powerful synergy that addresses both economic and integrity challenges simultaneously [2], [19].

## 1.2 Research Gap

The Sri Lankan pepper industry plays a vital role in the country's agricultural exports and farmer livelihoods. However, it faces persistent challenges, including high price volatility, opaque supply chains, and limited market transparency. Traditional forecasting methods like ARIMA and SARIMA have been applied to predict pepper prices, but these approaches largely rely on historical data and fail to account for external factors such as weather fluctuations, global demand, and market disruptions [10], [11].

Recent advancements in machine learning, including RF, XGBoost, LSTM and hybrid architectures such as TCN-XGBoost, have shown improved forecasting accuracy in agricultural price prediction [6], [12], [13]. Yet, many of these models act as “black boxes,” limiting stakeholder trust due to their lack of interpretability [14]. Although the TFT has been introduced as an interpretable deep learning model for multi-horizon forecasting [1], [4], its application to Sri Lanka's spice sector remains underexplored.

On the traceability front, blockchain has emerged as a transformative technology in agriculture, with studies demonstrating its ability to improve transparency, trust, and food safety in supply chains [2], [3], [5], [7], [8], [15]. Blockchain has been successfully applied in crops such as rice and coffee, often integrated with IoT and QR code technologies to provide tamper-proof product provenance [16], [17], [20]. However, its adoption in Sri Lanka's pepper industry is limited, with barriers such as cost, digital literacy, and infrastructure slowing implementation [18].

The following are the key research gaps identified from existing literature and practices:

1. **Limited integration of forecasting and traceability:** Existing studies treat price forecasting and supply chain transparency as separate domains. Few works attempt to combine interpretable forecasting with blockchain-enabled traceability for agriculture, and none have focused on Sri Lankan pepper [9], [19].
2. **Lack of interpretability in modern forecasting models:** While models like XGBoost, LSTM, and TCN-XGBoost achieve higher accuracy, they operate as black boxes, reducing trust among farmers and stakeholders [12], [14].

3. **Underutilization of blockchain in the pepper industry:** Blockchain traceability is growing in other crops, but Sri Lanka's pepper industry has yet to leverage this technology for trust and export competitiveness [2], [18].
4. **Absence of actionable farmer insights:** Prior studies do not provide integrated solutions that simultaneously empower farmers with predictive insights and verifiable supply chain transparency [15], [16].

To address these gaps, this research proposes the integration of TFT for interpretable price forecasting with blockchain-enabled supply chain traceability, delivering a holistic solution that strengthens both economic decision-making and trust in the Sri Lankan pepper industry.

| Feature / Gap                                                          | [10] | [12] | [14] | [2] | [18] | Proposed System |
|------------------------------------------------------------------------|------|------|------|-----|------|-----------------|
| Consider external factors (weather, demand, currency)                  | ✗    | ✓    | ✓    | ✗   | ✗    | ✓               |
| Provides interpretable and transparent forecasts                       | ✗    | ✗    | ✓    | ✗   | ✗    | ✓               |
| Focus on Sri Lanka's pepper industry                                   | ✗    | ✗    | ✗    | ✗   | ✓    | ✓               |
| Blockchain-enabled supply chain transparency                           | ✗    | ✗    | ✗    | ✓   | ✗    | ✓               |
| Actionable farmer insights (recommendations, alerts, decision support) | ✗    | ✗    | ✗    | ✗   | ✗    | ✓               |

*Table 1: Research Gap*

### **1.3 Research Problem**

In Sri Lanka's pepper industry, farmers and exporters face significant challenges due to highly volatile and unpredictable market prices. The absence of reliable forecasting systems forces many smallholder farmers to sell their produce at unfavorable times, leading to unstable income and reduced profitability. Traditional statistical forecasting methods such as ARIMA or SARIMA rely mainly on past price trends and fail to incorporate critical external drivers like global market demand, weather changes, input costs, and exchange rate fluctuations. These models also lack interpretability, making it difficult for farmers and decision-makers to understand why a prediction was made, further reducing trust and adoption among stakeholders. This creates a clear need for an advanced, interpretable, and farmer-friendly forecasting solution that integrates diverse data sources and provides actionable market price predictions.

In addition to forecasting challenges, the industry suffers from limited transparency and traceability across the supply chain. Current processes do not provide exporters, buyers, or consumers with verifiable information about the origin, grading quality, disease status, or certification of pepper batches. This lack of visibility creates distrust between stakeholders and results in inconsistent pricing, farmer exploitation, and reduced competitiveness of Sri Lankan pepper in global markets. Conventional record-keeping systems are fragmented, paper-based, or centralized, making them vulnerable to manipulation and data loss. Without a standardized and tamper-proof system to ensure data integrity, buyers are unable to confidently verify authenticity and quality.

As global markets increasingly demand transparency, sustainability, and proof of origin in agricultural supply chains, Sri Lanka's pepper industry risks losing competitive advantage if these problems remain unresolved. The dual challenge of unpredictable market prices and non-transparent supply chains highlights the urgent need for an integrated solution that not only predicts market prices using interpretable AI but also ensures trust and accountability through blockchain-based data integrity. Addressing this problem will directly improve farmer income stability, strengthen buyer confidence, and elevate Sri Lanka's position in the international spice trade.

## 2. OBJECTIVES

### 2.1 Main Objective

The objective of this research is to develop an AI and IoT-powered intelligent decision support system for smart pepper farming in Sri Lanka. The system integrates multiple functions; yield prediction, disease detection, market price forecasting, post-harvest grading, and blockchain-based traceability into one farmer-friendly solution.

- **To predict yield:** Machine learning models trained on plants and environmental data will provide harvest estimates, supported by AR visualizations to help farmers plan resources effectively.
- **To detect diseases:** AI-based image recognition will identify early signs of crop diseases through mobile phone images, enabling timely interventions and reducing yield losses.
- **To forecast market price:** Advanced forecasting models will analyze past data and international trends to predict prices, helping farmers decide the best time to sell.
- **To grade the quality:** An AI and IoT-enabled system will grade pepper based on density, visual quality, and certification compliance, ensuring fair valuation and export readiness.
- **Blockchain Traceability:** Each pepper batch will have a digital record accessible via QR codes, ensuring transparency, authenticity, and buyer trust.

By uniting these components, the system aims to enhance productivity, transparency, and profitability for smallholder farmers, strengthening the competitiveness and global reputation of Sri Lanka's pepper industry.

## **2.2 Specific Objectives**

### **Develop the market price forecasting module**

- Use advanced models like the TFT to predict pepper prices by integrating historical prices, weather, global demand, and exchange rates.

### **Provide interpretable forecasts and smart recommendations**

- Generate price forecasts with explanations of key drivers and deliver actionable advice such as “sell now” or “wait” to maximize farmer and exporter profits.

### **Implement blockchain enabled traceability**

- Record outputs from all system modules (yield, grading, and price) on a permissioned blockchain to ensure data integrity and security.

### **Enable QR code-based verification**

- Assign each pepper batch a unique QR code linked to blockchain records, allowing buyers, exporters, and consumers to verify product origin, quality, and certification.

### **Enhance trust and transparency in the supply chain**

- Strengthen farmer-to-buyer confidence by reducing fraud, ensuring fair pricing, and improving Sri Lanka’s competitiveness in the global pepper market.

### **3. METHODOLOGY**

#### **3.1 System Overview**

The proposed system is an integrated AI and IoT-powered intelligent decision support platform for smart pepper farming, designed to optimize cultivation, disease management, pricing decisions, and post-harvest quality grading. It leverages computer vision, deep learning, IoT sensors, AR visualization, advanced forecasting models, and blockchain to provide farmers and stakeholders with actionable insights, transparency, and improved profitability.

- **Yield Prediction and Visualization**

This component uses CNN based image analysis to predict the yield of each pepper plant from early-stage images of pepper cones. The predicted yield is then visualized using an AR enabled mobile application, allowing farmers to interactively view expected yield per plant. Additionally, spatial heatmaps are generated from plantation images, showing growth intensity and distribution across the field.

- **Disease Detection and Health Monitoring**

The system enables early disease detection using deep learning models applied to leaf images captured by farmers. It provides localized treatment recommendations, including organic and chemical solutions. Furthermore, NDVI based health assessments detect plant stress before visible symptoms, helping farmers take preventive actions. Smart notifications alert farmers to regional disease outbreaks and provide daily plant health tips for better crop management.

- **Market Price Forecasting and Blockchain Traceability**

This module employs TFT for multi-variable time-series forecasting, predicting weekly pepper prices by considering weather, production, and market factors. It offers “sell” or “wait” recommendations with confidence scores to guide farmers’ marketing strategies. Additionally, all critical data yield, disease status, price forecasts, and quality grades are stored on a blockchain to ensure tamper-proof traceability. A QR code linked to each pepper batch provides a transparent product history for consumers and exporters.

- **Post-Harvest Quality Grading**

The system integrates an IoT-based device (ESP32 with load cell and moisture sensor) to measure bulk density and moisture content of pepper samples. Combined with AI based image processing, the module evaluates visual quality factors such as color uniformity, mold presence, and adulteration. A multi-parameter scoring system classifies pepper into Grade A, B, or C, aligned with national and international standards. Farmers receive downloadable quality reports, and high-quality batches trigger buyer notifications for better pricing opportunities.

### 3.1.1 System Architecture

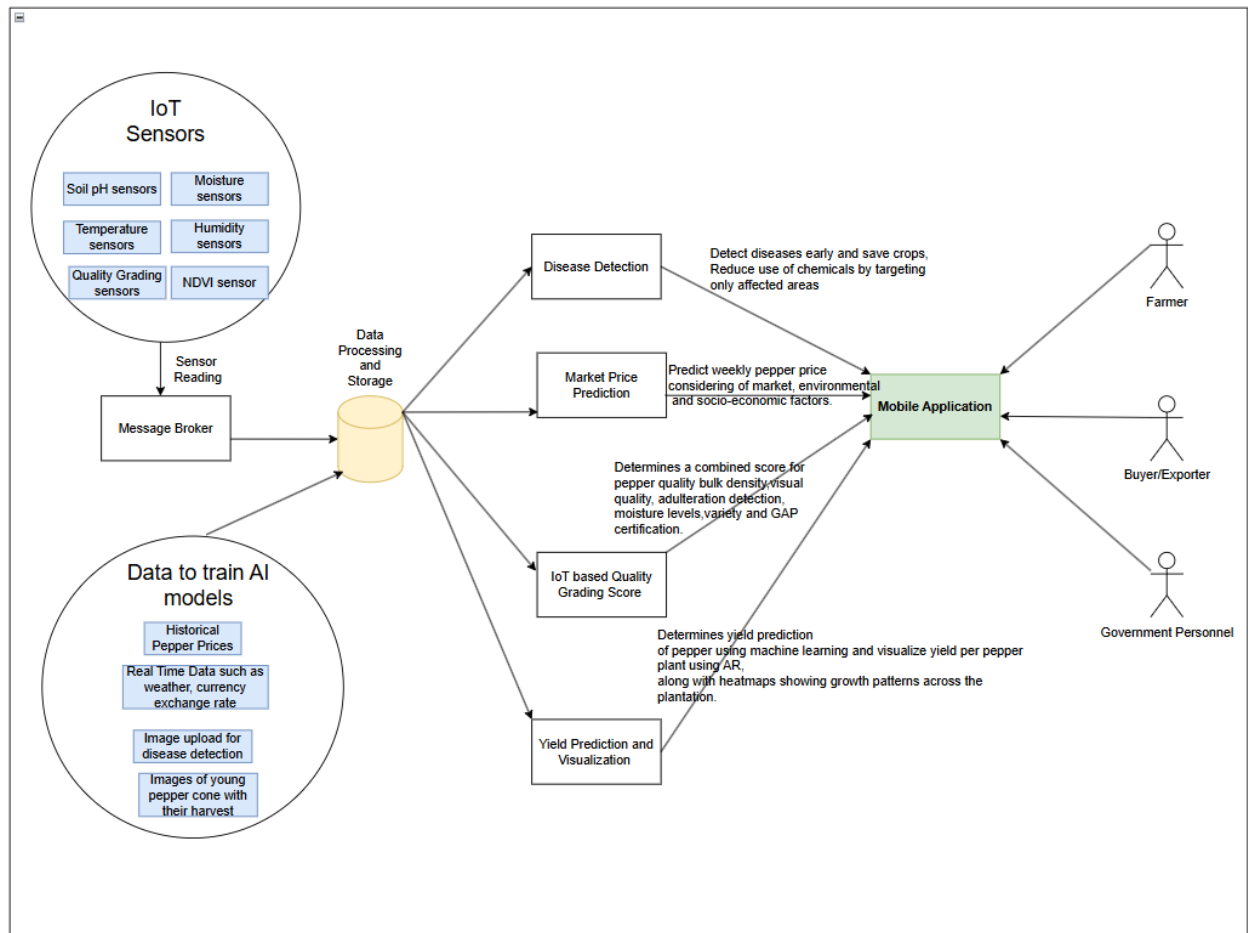


Figure 1: System Overview Diagram

The proposed system is designed as a modular AI and IoT-driven platform that integrates data collection, intelligent analytics, visualization, decision support, and blockchain-based traceability. It supports pepper farmers in Sri Lanka by predicting yield, detecting diseases early, forecasting market prices, grading post-harvest quality, and ensuring transparent supply chain tracking. The key Architectural Layers are,

#### 1. Data Acquisition Layer

- IoT Devices & Sensors

ESP32-based device with load cell and moisture sensor for bulk density and moisture measurement. Handheld NDVI sensor / modified camera for vegetation health detection.

- Image Capture

Farmers upload images of pepper cones (for yield prediction) and leaves (for disease detection) via the mobile app.

- External Data Sources

Weather APIs, currency exchange rates, and historical price databases for price forecasting.

## **2. Data Processing & AI Analytics Layer**

- Yield Prediction Module

Uses CNN based image analysis of young pepper cones to predict per-plant yield. Generates AR visualizations and heatmaps of growth patterns.

- Disease Detection Module

DL (CNN and NDVI) to detect disease symptoms early. Predicts disease progression and provides treatment recommendations.

- Market Price Forecasting Module

Uses TFT for multi-variable time-series forecasting. Consider historical prices, weather, production trends, and currency rates.

- Post-Harvest Quality Grading Module

Combines IoT measurements (bulk density) with image-based visual grading. Assigns Grade A/B/C based on combined score.

- Blockchain Traceability

Records key outputs (yield, quality score, price forecasts) in a permissioned blockchain. Generates QR codes for each batch, enabling full farm-to-market transparency.

### **3. Application & Visualization Layer**

- **Mobile App**

Farmers can capture images, view AR yield predictions, and receive disease alerts.  
Displays weekly price forecasts, sell/wait recommendations, and quality grading reports.  
Provides community features: success stories, tips, notifications.

- **Web Dashboard**

Used by agriculture officers, buyers, and exporters to monitor production trends, quality statistics, and traceability logs.

#### **3.1.2 Software Solution**

The software solution is a multi-layered architecture that leverages both on-device intelligence and cloud-edge hybrid processing to ensure accessibility even in rural areas with limited connectivity.

- **Mobile Application (Flutter Framework)**

Cross-platform support for Android/iOS, ensuring inclusiveness.

Enables image capture, AR visualization, and farmer interaction.

Offline caching and delayed synchronization ensure uninterrupted use.

- **Backend AI Services (Python, TensorFlow)**

Handles training, inference, and continuous model updates.

Deployed as microservices via REST APIs, enabling modular integration.

- **Visualization Layer**

Plantation-level heatmaps and analytics built using Plotly, Mapbox, or D3.js.

Supports both spatial and temporal analytics for multi-season trend analysis.

### 3.1.3 Commercialization

The system holds significant potential for technology transfer and agribusiness impact in Sri Lanka and beyond:

- **Target Users**

Smallholder pepper farmers (majority of Sri Lankan farmers).

Plantation cooperatives and spice exporters.

Government extension services and NGOs.

- **Business Model**

Free tier: Basic yield prediction by weight.

Premium tier: Advanced AR visualization, historical trend analysis, and detailed heatmaps via subscription.

B2B partnerships: Analytics and traceability solutions for exporters and certification bodies.

- **Value Proposition**

Farmers: Early insights, reduced uncertainty, better resource allocation, increased profitability.

Exporters: Reliable supply chain with predictable volumes.

Government/NGOs: Macro level crop monitoring and policy planning.

- **Scalability and Adaptability**

The architecture is cropping agnostic and can be extended to other spices (cardamom, cinnamon, nutmeg, etc.).

Potential for regional/global adoption in spice exporting economies.

## **4. PROJECT REQUIREMENTS**

### **4.1 Functional Requirements**

#### **1. Data Collection**

- Collect historical pepper price data (domestic and export).
- Collect related external factors such as weather, exchange rates, and export volumes.

#### **2. Price Forecasting**

- Process multi-source data to predict weekly pepper prices using TFT.
- Provide interpretable outputs showing key factors influencing the forecast.

#### **3. Recommendation & Alerts**

- Recommend “sell” or “wait” based on forecasted trends.
- Send alerts when favorable selling conditions are detected.

#### **4. Blockchain Recording**

- Store yield prediction, grading results, and price forecasts on a permission blockchain.

#### **5. QR Code Generation**

- Generate a unique QR code for each pepper batch linked to its blockchain record.

#### **6. Traceability Access**

- Allow users to scan QR codes to view verified product history and market information.

#### **7. User Interface**

- Provide a mobile/web interface for farmers, exporters, and buyers to access forecasts, recommendations, and traceability data.

## **4.2 Non-Functional Requirements**

### **1. Performance**

- Price predictions shall be generated within a maximum of 5 seconds after input data is processed.

### **2. Scalability**

- The system shall handle increasing amounts of historical and real-time data without performance degradation.

### **3. Security**

- All data shall be encrypted during storage and transmission.
- Blockchain records shall be tamper-proof.

### **4. Reliability**

- The system should ensure 99% uptime for critical services like prediction and QR code access.

### **5. Usability**

- The user interface shall be intuitive and accessible to non-technical users.

### **6. Interoperability**

- The system should be compatible with standard APIs for integrating weather services, market data providers, and export databases.

### **7. Maintainability**

- The system shall allow for model retraining and blockchain node updates with minimal downtime.

### 4.3 User Requirements

- Farmers should be able to view weekly pepper price forecasts with clear explanations.
- Users should receive price alerts when favorable market conditions are forecast.
- Exporters and buyers should be able to verify product authenticity and quality via blockchain.
- A QR code system should allow easy access to pepper batch history (origin, grading, forecast data).
- The platform should be accessible via mobile and web interfaces with a simple, user-friendly design.

### 4.4 System Requirements

#### Hardware Requirements

##### **For farmers to install Mobile Application:**

- Smartphone
- Internet connection

##### **For developers of AI Model training:**

- Processor: Intel i7/ i5
- GPU: NVIDIA RTX 3060 (or higher) with CUDA support
- RAM: 16GB minimum
- Storage: 1TB SSD for dataset and model checkpoints.

## **Software Requirements**

### **For development of mobile application:**

- Visual Studio - IDE for development.
- Flutter SDK - Cross platform mobile application development.
- Dart - Main programming language.
- Git and GitHub for version controlling.

### **AI/ML Frameworks for AI Model Development:**

- Python for model training.
- Jupyter Notebook for experimentation, training and preprocessing.
- TFT is for advanced time-series model for multi-variable forecasting, providing both accurate predictions and interpretable insights on key influencing factors.

### **Backend / Data Management:**

- Flask / FastAPI - for model inference API.
- Firebase (Authentication, Firestore, Cloud Storage) for user accounts, image uploads, and real-time database.
- SQLite for offline image storage and prediction history.

### **Testing and Deployment:**

- Flutter Testing Framework for unit, widget, and integration testing.
- Firebase Test Lab – to test app on different devices.

## 5. GANTT CHART

|                                    | 2025 |      |      |     |     |     |     |     | 2026 |     |       |       |     |
|------------------------------------|------|------|------|-----|-----|-----|-----|-----|------|-----|-------|-------|-----|
|                                    | May  | June | July | Aug | Sep | Oct | Nov | Dec | Jan  | Feb | March | April | May |
| Research Topic Selection           |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Topic Evaluation Form              |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Feasibility Study                  |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Requirement Gathering              |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal                   |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal Presentation      |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal Documentation     |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Requirement Specification |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Designing                 |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Implementation            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Unit Testing                       |      |      |      |     |     |     |     |     |      |     |       |       |     |
| System Integration Testing         |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Status Document            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Progress Presentation 1            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Writing Research Paper             |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Testing Level 2                    |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Progress Presentation 2            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Demo Level 1                       |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Website Assesment                  |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Status Document 2          |      |      |      |     |     |     |     |     |      |     |       |       |     |
| LogBook                            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Final Presenation and Viva         |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Final Thesis                       |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Final Report                       |      |      |      |     |     |     |     |     |      |     |       |       |     |

Figure 2: Gantt Chart

## **6. DESCRIPTION OF PERSONAL AND FACILITIES**

- Facilitators:
  - Ms. Ishara Weerathunga (Supervisor) – Sri Lanka Institute of Information Technology (SLIIT)
  - Mr. Eishan Weerasinghe (Co-Supervisor) – Sri Lanka Institute of Information Technology (SLIIT)
  
- Facilities:
  - Datasets

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## APPENDICES

The screenshot displays the Turnitin Feedback Studio interface. The main document area shows the title "AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA", the ID "25-26J-479", and the author "Ridimahaliyadda A.K.". The document is a "Project Proposal Report" for a "B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering". The similarity score is 6%. The right sidebar shows a "Match Overview" with a list of matches:

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Figure 3: Turnitin Result